

More Complicated Logarithmic Equations (Homework)

Solve. Write the Roman numeral of each exponent rule that you use.

Answers

$$\textcircled{1} \log_2(3+5x)=1$$

$$x = -\frac{1}{5}$$

$$\textcircled{2} 3 = \log_5(4 + \frac{x}{3})$$

$$x = 363$$

$$\textcircled{3} \log_4(-6x-1)=0$$

$$x = -\frac{1}{3}$$

$$\textcircled{4} -4 = \log_3(\frac{x}{2}-1)$$

$$x = \frac{164}{81}$$

$$\textcircled{5} 8\log_9(7x+6)-10=-6$$

$$x = -\frac{3}{7}$$

$$\textcircled{6} 2 = \frac{\log_9(2-8x)}{2} + 1$$

$$x = -\frac{79}{8}$$

$$\textcircled{7} 18 = 15 + 4\log_{16}(3 - \frac{x}{10})$$

$$x = -50$$

$$\textcircled{8} \frac{\log_5(-9x+2)}{4} - 6 = -5\frac{3}{4}$$

$$x = -\frac{1}{3}$$

$$\textcircled{9} 8\log_7(3x-2) + \frac{1}{3} = \frac{2}{3}$$

$$x = \frac{\sqrt[24]{7} + 2}{3}$$

Solve without a calculator. Write the Roman numeral of each exponent rule or logarithm property that you use.

$$\textcircled{10} \log_3 6 + \log_3 x = 1$$

$$x = \frac{1}{2}$$

$$\textcircled{11} \log_2 x - \log_2 5 = 3$$

$$x = 40$$

$$\textcircled{12} \log_8 y + \log_8 7 = \frac{1}{3}$$

$$y = \frac{2}{7}$$

$$\textcircled{13} \log_4 b - \log_4 3 = -1$$

$$b = \frac{3}{4}$$

$$\textcircled{14} 2\log_9 7 + \log_9 x = 2$$

$$x = \frac{81}{49}$$

$$\textcircled{15} 3\log_2 y - \log_2 5 = 1$$

$$y = \sqrt[3]{10}$$

$$\textcircled{16} 2\log_5 x - 2\log_5 6 = 0$$

$$x = 6$$

$$\textcircled{17} 4\log_3 2 + 6\log_3 b = 4$$

$$b = \sqrt[6]{\frac{81}{16}}$$

$$\textcircled{18} 3\log_2 y + \log_2(6y) + \log_2 10 = 4$$

$$y = \sqrt[4]{\frac{4}{15}}$$

$$\textcircled{19} \log_5 16 - 5 \log_5 2 + \log_5 (2x) = 3$$

$$\boxed{x = 125}$$